Sample Paper 03

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

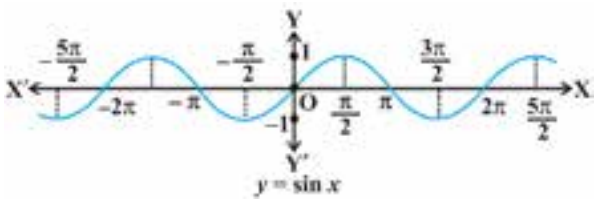
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. Which of the following points lies on the curve shown in the image?



(a) $\left(\frac{\pi}{2}, 0\right)$ (b) $\left(\frac{\pi}{2}, 1\right)$

(c) $(\pi, 1)$ (d) $(2\pi, -1)$

2. If $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = 3\hat{i} + 2\hat{j} - \hat{k}$, then the value of $(\vec{a} + 3\vec{b}).(2\vec{a} - \vec{b})$ is-

(a) 15 (b) 18

(c) -18 (d) -15

3. The maximum value of $\left(\frac{1}{x}\right)^{2x^2}$ is

(a) 1 (b) e

(c) $e^{1/e}$ (d) None of these

4. Let l_1, m_1, n_1 and l_2, m_2, n_2 be the direction cosines of two straight lines. Both the lines are perpendicular to each other, if

(a) $l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$

(b) $l_1 l_2 + m_1 m_2 + n_1 n_2 = 1$

(c) $\frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2}$

(d) $\frac{l_1}{l_2} + \frac{m_1}{m_2} + \frac{n_1}{n_2} = 0$

5. The direction cosines of the $y-$axis are

(a) (0, 0, 0) (b) (1, 0, 0)

(c) (0, 1, 0) (d) (0, 0, 1)

6. If $y = \log\cos x^2$, then $\frac{dy}{dx}$ at $x = \sqrt{\pi}$ has the value

(a) 1 (b) $\frac{\pi}{4}$

(c) 0 (d) $\sqrt{\pi}$

7. If $-1 < x < 1$, then $2\tan^{-1}x = ?$

(a) $\sin^{-1}\frac{2x}{1+x^2}$ (b) $\sin^{-1}\frac{2x}{1-x^2}$

(c) $\sin^{-1}\frac{1-x^2}{1+x^2}$ (d) $\sin^{-1}\frac{1+x^2}{1-x^2}$

8. $f : A \to B$ will be an onto function if-

(a) $f(A) \subset B$ (b) $f(A) = B$

(c) $f(A) \supset B$ (d) $f(A) \neq B$

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9. Feasible region is the set of points which satisfy:

(a) The objective function

(b) Some of the given constraints

(c) All the given constraints

(d) None of these

10. At time t distance of a particle moving in a straight line is given by $s = 4t^2 + 2t$. The acceleration when $t = \frac{1}{2}$, is

(a) 4 (b) 6

(c) 8 (d) 10

11. Solution of $\lim_{n\to\infty}\left[\frac{e^{1/n} + e^{2/n} + e^{3/n} + ... + e^{r/n}}{n}\right]$

(a) $1 - e$ (b) $e - 1$

(c) e (d) 1

12. $\int_0^1 \frac{dx}{1+x^2} = ?$

(a) $\frac{-\pi}{4}$ (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$ (d) $\frac{-\pi}{2}$

13. If $2\begin{bmatrix}3 & 4\\ 5 & x\end{bmatrix}+\begin{bmatrix}1 & y\\ 0 & 1\end{bmatrix}=\begin{bmatrix}7 & 0\\ 10 & 5\end{bmatrix}$, then-

(a) $(x = -2, y = 8)$

(b) $(x = 2, y = -8)$

(c) $(x = 3, y = -6)$

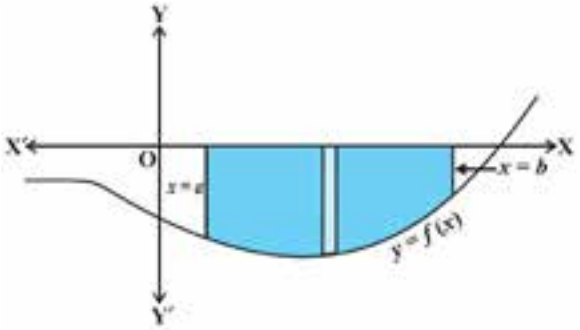
(d) $(x = -3, y = 6)$

14. The value of x when $\begin{bmatrix}x & 15\\ 4 & 4\end{bmatrix} = 0$ is-

(a) 15 (b) –15

(c) 4 (d) $4x$

15. In Fig., the shaded region lies between which of the following?



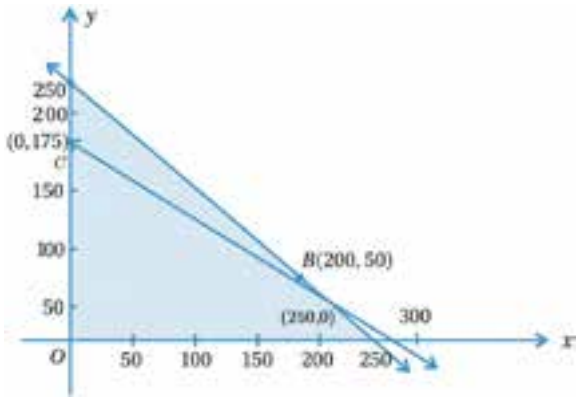
(a) The curve y=f (x), the X-axis, and the vertical lines x=a and x=b

(b) The curve x=f (y), the Y-axis, and the horizontal lines y=a and y=b

(c) The curve y=f (x), the Y-axis, and the horizontal lines y=c and y=d

(d) The curve x=f (y), the X-axis, and the vertical lines x=a and x=b

16. The corner points of the bounded feasible region of an LPP are O(0,0), A(250,0), B(200,50), and C(0,175). If the maximum value of the objective function $Z = 2ax +$ by occurs at the points A(250,0) and B(200,50), then the relation between a and b is to be determined.



(a) 2a=b (b) 2a=3b

(c) a=b (d) a=2b

17. [Standard result] $\int \sqrt{1+\cos 2x}\, dx =$

(a) $\sqrt{2}\cos x + c$ (b) $\sqrt{2}\sin x + c$

(c) $-\cos x - \sin x + c$ (d) $\sqrt{2}\sin\frac{x}{2} + c$

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18. Integrating factor of the differential equation $\frac{dy}{dx} + y\sec x = \tan x$ is-

(a) $\sec x + \tan x$ (b) $\sec x - \tan x$

(c) $\sec x$ (d) $\tan x \sec x$

19. **Assertion :** If $P(E_1) = 0.3$ and $P(\overline{E_2}) = 0.6$ then $P(E_1 \cup E_2) = 0.7$

Reason : $P(\overline{E_2}) = 1 - P(E_2)$

(a) Both assertion and reason are true and reason is the correct explanation of assertion.

(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) Assertion is true, but reason is false.

(d) Assertion is false, but reason is true.

20. **Assertion :** If $y = \sin x^3$, then $\frac{dy}{dx} = \cos x^3 \cdot 3x^2$.

Reason: $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$.

(a) Both assertion and reason are true and reason is the correct explanation of assertion.

(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) Assertion is true, but reason is false.

(d) Assertion is false, but reason is true.

SECTION - B

This section comprises of very short answer-type questions (VSA) of 2 marks each.

21. Write the vector equation of a line passing through point $(1, -1, 2)$ and parallel to the line whose equation is $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z+1}{-2}$.

22. Two groups are computing for the positions of the Board of Directors of a corporation. The probabilities that the first and second groups will win are 0.6 and 0.4 respectively. Further, if the first group wins, the probability of introducing a new product introduced way by the second group.

23. Find the differential equation representing the family of curves $y = ae^{bx+5}$, where a and b are arbitrary constants.

24. Write the value of $\int_0^1 \frac{e^x}{1+e^{2x}} dx$.

25. Write a unit vector in the direction of the sum of the vectors $\vec{a} = 2\hat{i} + 2\hat{j} - 5\hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - 7\hat{k}$.

OR

If $\vec{a} = \hat{i} - \hat{j} + 7\hat{k}$ and $\vec{b} = 5\hat{i} - \hat{j} + \lambda\hat{k}$, then find the value of λ, so that $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are perpendicular vectors.

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. Evaluate $\int \frac{x^2 + x + 1}{(x^2+1)(x+2)} dx$.

27. Given $A = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$, compute A^{-1} and show that $2A^{-1} = 9I - A$.

OR

If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$ be such that $A^{-1} = kA$, then find the value of k.

28. If $A = \{1, 2, 3\}, B = \{4, 5, 6, 7\}$ and $f = \{(1, 4), (2, 5), (3, 6)\}$ is a function from A to B. State whether f is one-one or not.

29. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 4\hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{c} = \hat{i} - 2\hat{j} + \hat{k}$. Find a vector of magnitude 6 units, which is parallel to the vector $2\vec{a} - \vec{b} + 3\vec{c}$.

30. Evaluate $\int_0^{\pi/2} x^2 \sin x \, dx$.

OR

Evaluate $\int_{\pi/4}^{\pi/2} \cos 2x \cdot \log(\sin x) \, dx$.

31. Solve for x, $\cos^{-1} x + \sin^{-1}\left(\frac{x}{2}\right) = \frac{\pi}{6}$.

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

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32. If $A=\begin{bmatrix}1 & -1\\ 2 & -1\end{bmatrix}$, $B=\begin{bmatrix}a & 1\\ b & -1\end{bmatrix}$ and $(A+B)^2=A^2+B^2$, then find the values of a and b.

OR

Find the adjoint of the matrix $A=\begin{bmatrix}-1 & -2 & -2\\ 2 & 1 & -2\\ 2 & -2 & 1\end{bmatrix}$ and

hence show that $A(adj\ A)=|A|I_3$.

33. If $x=\cos t+\log\tan\left(\frac{t}{2}\right)$, $y=\sin t$, then find the values of $\frac{d^2y}{dt^2}$ and $\frac{d^2y}{dx^2}$ at $t=\frac{\pi}{4}$.

OR

Find the values of a and b, if the function f defined by $f(x)=\begin{cases}x^2+3x+a, & x\le 1\\ bx+2, & x>1\end{cases}$ is differentiable at $x=1$.

34. Let $\vec{a}=\hat{i}+4\hat{j}+2\hat{k}$, $\vec{b}=3\hat{i}-2\hat{j}+7\hat{k}$ and $\vec{c}=2\hat{i}-\hat{j}+4\hat{k}$. Find a vector $\vec{p}$, which is perpendicular to both $\vec{a}$ and $\vec{b}$ and $\vec{p}\cdot\vec{c}=18$.

35. Find the particular solution of the differential equation satisfying the given condition.

$$x^2\,dy+(xy+y^2)\,dx=0,\text{ when } y(1)=1$$

OR

Find the particular solution of the differential equation

$$x\frac{dy}{dx}-y+x\,\mathrm{cosec}\left(\frac{y}{x}\right)=0$$

SECTION - E

Case study based questions are compulsory.

36. An insurance company believes that people can be divided into two classes: those who are accident prone and those who are not. The company's statistics show that an accident-prone person will have an accident at sometime within a fixed one-year period with probability 0.6, whereas this probability is 0.2 for a person who is not accident prone. The company knows that 20% of the population is accident prone.



On the basis of above information, answer the following questions.

(i) What is the probability that a new policyholder will have an accident within a year of purchasing a policy?

(ii) Suppose that a new policy holder has an accident within a year of purchasing a policy. What is the probability that he or she is accident prone?

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37. Chemical reaction, a process in which one or more substances, the reactants, are converted to one or more different substances, the products. Substances are either chemical elements or compounds. A chemical reaction rearranges the constituent atoms of the reactants to create different substances as products.



In a certain chemical reaction, a substance is converted into another substance at a rate proportional to the square of the amount of the first substance present at any time t. Initially $(t = 0)$ 50 g of the first substance was present; 1 hr later, only 10 g of it remained.

(i) Find an expression that gives the amount of the first substance present at any time t.

(ii) What is the amount present after 2 hr?

38. Vitamins are nutritional substances which you need in small amounts in your diet. Vitamins A and E are fat-soluble vitamins, meaning they're stored in your body's fat cells, but they need to have their levels topped up regularly. Vitamin C is a water-soluble vitamin found in citrus and other fruits and vegetables, and also sold as a dietary supplement. It is used to prevent and treat scurvy. Vitamin C is an essential nutrient involved in the repair of tissue, the formation of collagen, and the enzymatic production of certain neurotransmitters.





A dietician wishes to mix two types of foods in such a way that the vitamin contents of mixture contains atleast 8 units of vitamin A and 10 units of vitamin C. Food I contains 2 units per kg of vitamin A and 1 unit per kg of vitamin C, while food II contains 1 unit per kg of vitamin A an 2 units per kg of vitamin C. It costs Rs. 30 per kg to purchase food I and Rs. 42 per kg to purchase food II.

(i) Formulate above as an LPP and solve it graphically.

(ii) Find the minimum cost of such a mixture.

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